Wesley Wong

High Schooler in Embedded Autonomous Systems
San Jose, CA | linkedin.com/in/wesley-wong28 | github.com/Wesley28w | wesleycwong.com

Education

Los Gatos High School Expected May 2029
GPA: 4.5 weighted / 4.0 unweighted | Focus in Computer Science, Robotics, ML, and Embedded Systems *Los Gatos, CA*

Experience

Independent Researcher and Solo Author Dec. 2025 – Present
Robotics, Reinforcement Learning, and Embedded Autonomy *San Jose, CA*

- Sole-authored research on reinforcement-learning curricula for robotic manipulation. Accepted to the NeurIPS 2026 PhysUnderstand Workshop.
- Endorsed to submit in arXiv cs.LG with a TMLR submission pending. Continuing research in multimodal robotics, spiking neural networks, and machine learning.

Co-Founder, Co-President, and Software/AI Director Jan. 2026 – Present
Los Gatos CubeSat *Bay Area, CA*

- Co-founded and lead software/AI for a 20-member CubeSat team. Programmed an eight-process Linux OBC and integrated the complete flight system across software and hardware.
- Raised $10,000+ in initial funding. Develop STM32/FreeRTOS avionics and spiking-neural-network attitude control on neuromorphic hardware.

Director of Software May 2025 – May 2026
FRC Team 972 Iron Claw Robotics *Bay Area, CA*

- Promoted to Director of Software in under six months, then led software for a 50-member FIRST Robotics Competition team. Helped earn three awards, including at the state championship, and reach the Houston World Championship Division Finals.

Bay Area Ambassador Aug. 2026 – Present
Exea Labs *Bay Area, CA*

- Represent the Bay Area for a network of 800+ researchers across 30 countries, backed by AMD, Siemens, and Microsoft.
- Connect students with compute resources, publication guidance, and research feedback. Plan workshops that help students launch research projects.

Projects

TriBot | *Onshape, Simulink, CFD, STM32, C* June 2026 – Present

- Designing a multimodal robot for ground, aerial, and aquatic operation. Model mechanical systems in Onshape, analyze flight and water dynamics with Simulink and CFD, and develop STM32 firmware in C.

Synopsys Science Fair: GPT-Controlled RL Pipeline | *Python, GPT, Isaac Lab* Dec. 2025 – Apr. 2026

- Built an AI-powered engineering pipeline that decomposed robotics objectives into structured tasks and used multi-stage validation to generate and test RL components. Placed third in category at SCVSEFA.

Self-Hosted GPU Cluster | *Vast.ai, Google Cloud Run, Redis, vLLM* Aug. 2025 – Dec. 2025

- Built an on-demand AI inference platform with automatic GPU routing, low-latency vLLM serving, Redis caching, and a Cloud Run control layer.

AI From Scratch | *Python, NumPy, Linear Algebra* Sep. 2026 – Present

- Built linear and logistic regression, KNN, PCA/SVD, K-means, and decision trees from first principles as part of a self-designed 12-project curriculum focused on mathematical derivation and implementation.

Awards and Honors

NeurIPS 2026 PhysUnderstand Workshop Paper Acceptance | arXiv cs.LG Endorsement | SCVSEFA Category Third Place
FIRST World Championship Division Finalist | Three FIRST Robotics Awards | Los Gatos High School Researcher Award

Technical Skills

Languages: Python, C, Embedded C, Swift, Java
Machine Learning and Robotics: PyTorch, NumPy, NVIDIA Isaac Lab, reinforcement learning, YOLO, OpenCV, transformers, SNNs
Embedded and Simulation: Linux, FreeRTOS, Raspberry Pi, STM32, I2C/SPI/UART, Onshape, Simulink, Simscape Multibody, CFD
Cloud and Developer Tools: Git/GitHub, CMake, Docker, Cloud Run, Redis, vLLM, Vast.ai, LaTeX